

Simplifying assumptions were used in early research, as the purpose was to provide a more realistic assessment of sustainable spending than found when assuming a fixed average investment return. But these studies subsequently took on a life of their own. The 4 percent rule has been widely adopted by the popular press and financial planners as an appropriate rule of thumb for retirees. The 4 percent rule is so widely ingrained in the culture as a universal standard that people commonly think it must apply to any retirement length. Indeed, people are often surprised to learn it is specifically designed for a thirty-year retirement. The 4 percent rule wasn't necessarily meant to apply to eighty-five-year-olds, nor can it be safely used by early retirees who leave the workforce by age forty.

The basic philosophy and assumptions behind the 4 percent rule include that the objective is to meet an overall lifestyle spending goal. Retirees are assumed to desire smooth spending, but they also have an appetite for market volatility. Retirees do not voluntarily reduce spending as they age or adjust withdrawals in response to realized financial market returns. Withdrawals are constant, inflation-adjusted amounts.

Failure in retirement is defined as not meeting the overall spending goal for the full assumed retirement time horizon. The underlying objective is to keep the failure rate (the probability of portfolio depletion) at a reasonably low level. For market risk management, retirees use a diversified portfolio focused on total returns with spending from income and principal. Historically, 50–75 percent stocks maximizes the probability of plan success. For longevity risk management, retirees assume a planning horizon sufficiently beyond life expectancy. The 4 percent rule is calibrated specifically to a thirty-year horizon. For management of spending shocks, retirees focus on precautionary savings. As the 4 percent rule is calibrated to be sustainable in the worst case from US history, it otherwise preserves assets that could potentially be deployed for contingencies in other cases.

Basic assumptions also include that retirees earn the precise underlying investment returns net of any fees for a fixed asset allocation with annual portfolio rebalancing. The investment portfolio is either tax deferred or tax free. The two financial assets are large-capitalization stocks (S&P 500) and intermediate-term US government bonds. Finally, the 4 percent rule